

LIVELINE

On-Device Live Captioning & Translation for Snapdragon PCs

THE PROBLEM

Real-time captions still depend on the cloud

Privacy risk

Meeting and classroom audio is streamed to third-party servers to get live captions.

No offline access

Rural schools, field teams, and low-bandwidth regions in India lose captioning exactly when they need it most.

Cost & latency

Cloud speech APIs add round-trip lag and per-minute costs that don't scale for continuous use.

THE SOLUTION

A fully offline captioning assistant, on your Snapdragon PC

LiveLine transcribes speech in real time and optionally translates it, running entirely on the Snapdragon NPU — no internet, no cloud, no audio ever leaves the device.

Offline-first

Works with zero connectivity — proven live in airplane mode.

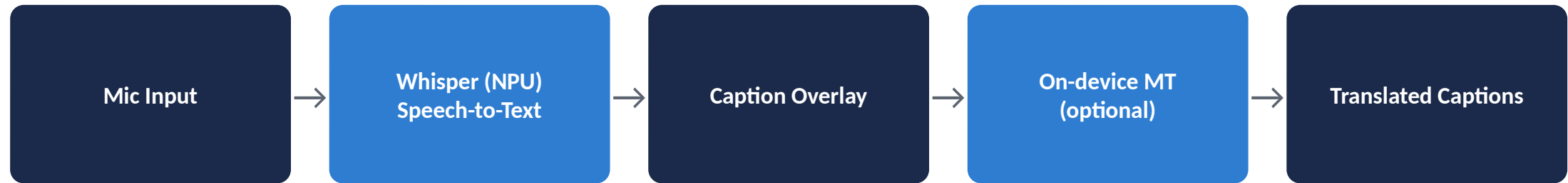
Private by design

All inference happens on-device. Nothing is uploaded.

Built for accessibility

Serves hearing-impaired users, non-native speakers, and low-bandwidth classrooms.

How it works, end to end



Runtime: Qualcomm AI Hub Whisper (QNN / ONNX Runtime) accelerated on the Snapdragon NPU. Translation layer runs on lightweight CPU-side MT model.

On-device isn't a constraint — it's the advantage

Latency

Whisper encoder runs in ~21ms on NPU — captions feel instant, not laggy.

Battery life

NPU inference draws far less power than sustained cloud streaming + radio use.

Privacy

No audio or transcript ever leaves the laptop — critical for meetings, interviews, classrooms.

Accessibility reach

No dependency on connectivity quality — works identically in a city office or a rural school.

Built on Qualcomm AI Hub models

- Speech-to-text: Whisper-Tiny / Whisper-Small-Quantized (Qualcomm AI Hub), deployed via pre-compiled QNN ONNX binaries for the target Snapdragon chipset.
- Inference runtime: ONNX Runtime with the QNN execution provider — NPU-accelerated, no cloud round-trip.
- Translation: lightweight on-device MT model, CPU-side, kept optional so the core offline story stays fast.
- Interface: always-on-top caption overlay with language selection and transcript export.
- Benchmarking: NPU vs. CPU inference time measured directly on the target Snapdragon-powered HP PC.

Proof, purpose, and what's next

Demo

- Live offline captioning, shown in airplane mode
- Real meeting / lecture scenario
- Side-by-side NPU vs CPU latency

Impact

- Hearing-impaired users & non-native speakers
- Low-bandwidth classrooms and field teams
- Zero-cost, privacy-preserving captioning

Roadmap

- Additional Indian language support
- Sign-language recognition module
- Mobile / Snapdragon handset port